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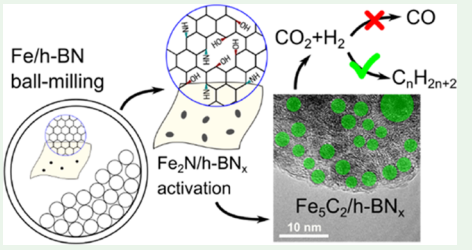
Ball-Milled Processed, Selective Fe/*h*-BN Nanocatalysts for CO_2 Hydrogenation

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ABSTRACT: Fe/*h*-BN nanocatalysts were obtained by a combination of precipitation and high-energy ball-milling (HEBM) techniques. Mechanical treatment at a rate of 300 and 500 rpm induced defects in the *h*-BN lattice and led to the formation of iron nitride nanoparticles when processing at a higher ball-milling mode. Lewis base centers formed as a result of the interaction of *h*-BN surface defects with water vapor from ambient air led to an increase in the activity of the Fe/*h*-BN catalyst in the CO_2 hydrogenation. Iron carbides, 2−4 nm in size, formed during the activation stage in the Fe/*h*-BN nanocomposites after HEBM increased the selectivity toward hydrocarbon formation. Calculations based on the density functional theory showed a significant weakening of the CO bond on the surface of iron carbide and suggested the iron nitride → iron carbide transformation as the most energetically favorable route.



KEYWORDS: *hexagonal boron nitride, ball-milling, lattice defects, CO_2 hydrogenation, phase transitions*

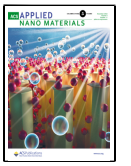
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INTRODUCTION

Rapid population growth led to environmental problems challenging scientific and industrial societies in developing ecologically friendly technologies. Currently, one of the main problems is CO_2 pollution of the atmospheric air, which has led to an increase in the global average temperature.[1] Modern CO_2 abatement strategies are based on reducing emissions through alternative energy sources and improving energy efficiency in combination with clean carbon dioxide removal (CDR) technologies.[2] To achieve the target of 2 °C average temperature reduction specified in the Paris Agreement, a global carbon capture scheme has been proposed, in which CO_2 utilization and storage will be implemented through sedimentary basin storage and CO_2 enhanced oil recovery.[3] Another promising CDR technology is the chemical conversion of CO_2 into valuable products with "green" hydrogen that partially offset the cost of carbon dioxide utilization.[4] Among them, carbon monoxide and methane are considered as useful CO_2 hydrogenation products for various applications. For example, methane can be used as intermediate energy storage for excess energy generation from solar and wind power plants.[5,6] Carbon monoxide can be used to produce either chemicals or fuels through the Fischer−Tropsch synthesis.[7]

Thermocatalytic hydrogenation of CO_2 to hydrocarbons (HCs) is being intensively studied by various scientific groups. A wide variety of catalytic systems including noble metals[5,8] and cheaper transition metals have been tested.[9−12] Among them, nanoparticles of ruthenium, cobalt, and nickel catalysts show the highest activity; however, these materials do not possess the required selectivity toward HCs.[13−15] In this regard, iron can be considered as a promising alternative catalyst. Currently, the greatest efforts in the study of iron-based catalysts are directed to the search for promoters and carriers capable of activating reaction molecules and ensuring optimal interaction with the active phases of catalysts.[16,17] Among the various carriers, oxide systems are paid the most attention. Nanostructured materials based on Al_2O_3,[18] zeolites,[19,20] TiO_2,[21] and CeO_2[22] were employed as the support materials. The multifunctional nature of catalysts, comprising active sites capable of activating different reaction intermediates, is noted as the crucial factor to attain high selectivity toward HC formation.[23] The main drawback of oxide carriers is associated with their physicochemical properties. For example, the high hydrophilicity of oxide supports leads to the poisoning of their surface with water forming during hydrogenation. Moreover, in the case of exothermal CO_2 reduction, low thermal conductivity of oxides can cause local temperature gradients.[24] Thus, the develop-

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ment of new catalytic supports for CO_2 hydrogenation is of great interest.

Hexagonal boron nitride (*h*-BN) nanopowders are promising candidates as an alternative carrier material because of their high specific surface area, relatively high chemical inertness, superior oxidation resistance, high thermal conductivity, and suspension stability allowing for uniform distribution of active nanoparticles on the support surface. Previously, we have shown that *h*-BN-supported catalysts can be successfully prepared by using a classical wet chemistry process and utilized in various oxidation reactions.[25,26] However, information about *h*-BN-based hydrogenation catalysts is rather scarce. The important role of *h*-BN lattice defects in the Fischer–Tropsch synthesis was noted.[27] Defects not only contribute to particle size reduction but also participate in the catalytic process providing new adsorption sites for reactants. It was shown that B and N vacancies are Lewis base and acid centers capable of catalyzing the nitroaldol reaction and glucose transformation.[28] These centers are known to activate relatively inert CO_2 molecules;[29,30] however, the effect of *h*-BN defects on the activity of iron-based catalysts in CO_2 hydrogenation has not been studied previously.

Lattice defects can be created in various ways, both during the *h*-BN synthesis[31] and as a result of subsequent post-treatment by chemical[32] and physical methods.[33] Mechanical processing using a ball mill is a promising method that has proven useful in catalyst preparation. Mechanical activation leads to a high density of defects as well as the formation of new metastable phases not achievable by traditional methods.[34] It is important to note that the ball milling of Fe/*h*-BN nanohybrids can cause not only the formation of acid and basic sites on the *h*-BN surface, but also induce defects in the Fe active phase, thereby enhancing the interaction between the deposited Fe and the carrier. This can not only significantly affect the catalytic properties, but also provide control over the activity and selectivity of Fe/*h*-BN catalysts in the CO_2 hydrogenation reaction.

In this work, for the first time, the influence of ball-milling-assisted mechanical treatment of Fe/*h*-BN nanohybrids on their catalytic activity and selectivity in the CO_2 hydrogenation reaction was studied.

RESULTS

To study the effect of ball milling on the catalytic activity and selectivity of Fe/*h*-BN nanohybrids in the CO_2 hydrogenation reaction, a series of samples was synthesized by wet chemical precipitation followed by high-temperature annealing in a hydrogen atmosphere and ball milling at different speeds, as shown in Figure 1.

Phase Composition of Fe/*h*-BN Nanohybrids. X-ray diffraction (XRD) patterns of the Fe/*h*-BN nanohybrids are presented in Figure 2a. After reduction, the sample consists of *h*-BN, metallic iron, and ferrous oxide phases, which are evidenced by the corresponding peaks at 39.8 and 86.3 for *h*-BN (Crystallography open database (COD) card No. 9008997), 68.6 and 106.1 for Fe (COD No. 9008536), and 55.1, 64.2, 98.7, 125.3, and 133.0 2θ for FeO (COD No. 9008636).[35] After ball milling at a speed of 300 rpm, the peaks of ferrous oxide noticeably diminished while the intensity of the iron peak increased and broadened. New peaks observed at 45.5 and 53.3 2θ can be ascribed to the magnetite phase (COD No. 9009768). Note that the full width at half-maximum (FWHM) value of the (002) *h*-BN peak significantly increased and the peak shifted to a lower 2θ value. This can be attributed to an increase in defectiveness of the BN lattice after the mechanical treatment.[36] The formation of boron oxide-containing phases was not observed even after the sample storage in ambient air for 1 month. Further increase in ball milling speed resulted in the disappearance of ferrous oxide and metallic iron phases. The new peaks at 57.1, 62.1, 65.6, 89.7, 111.38, and 131.7 appeared and were attributed to the ζ-Fe_2N phase.[37] Further broadening of the (002) *h*-BN peak is also noted.

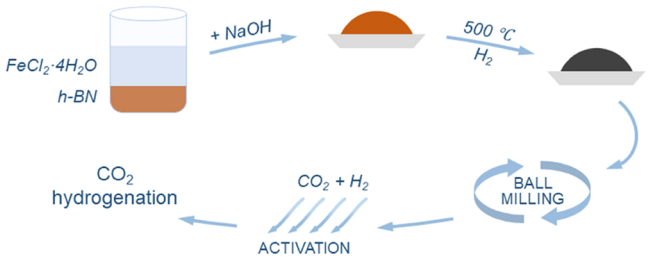


Figure 1. Schematics of the synthesis of Fe/*h*-BN nanohybrids.

Because the activation of catalysts affects their chemical composition and microstructure, the samples after activation in a CO_2/H_2 mixture were studied by XRD (Figure 2b). It can be seen that the activation of the Fe/*h*-BN sample led to a decrease in the intensity of FeO peaks with the simultaneous appearance of peaks corresponding to the magnetite phase. A similar observation was made in the case of the Fe/*h*-BN-300 sample, in which the relative intensity of the Fe_3O_4 peaks increased compared to the Fe peaks. Thus, the Fe/*h*-BN and Fe/*h*-BN-300 nanohybrids underwent slight oxidation under activation. In contrast, the Fe/*h*-BN-500 sample showed the formation of new carbide phases, Fe_3C and Fe_5C_2, as evidenced by the appearance of new peaks in the corresponding XRD pattern.

Surface Chemical State. Fourier-transform infrared (FTIR) spectra of the Fe/*h*-BN samples are presented in Figure 2c,d. The FTIR spectrum of the as-synthesized Fe/*h*-BN nanohybrid showed characteristic peaks of *h*-BN at 1364 cm^{-1}, corresponding to in-plane B–N bond stretching, and at 797 cm^{-1} due to out-of-plane B–N–B bending vibrations.[38] The peak at 552 cm^{-1} corresponds to the Fe–O bond vibration.[39] The minor peak at 681 cm^{-1}, which is attributed to the B–O–B ring bending,[40] indicates the presence of boron oxide not detected by XRD analysis. This can be either due to its low content or due to the presence of oxygen atoms as impurities on the BN surface. After ball milling, the broadening of *h*-BN peaks and a small redshift of the peak at 797 cm^{-1} are observed and attributed to the BN lattice defects induced by ball milling.[41,42] For the ball-milled sample at high speed, increased absorbance in the high-energy part of spectrum is observed. The bands in the range of 3500–3000 cm^{-1} correspond to the vibration of O–H and N–H bonds[43] indicating surface functionalization.

After activation, the FTIR spectra of the Fe/*h*-BN and Fe/*h*-BN-300 samples revealed an increased intensity of the 552 cm^{-1} peak corresponding to the vibration of Fe–O bonds (Figure 2d). This, in accordance with the results of XRD analysis, indicates the iron oxidation. In contrast, the FTIR spectrum of the Fe/*h*-BN-500 sample showed no Fe–O peak, but a new peak appeared at 1225 cm^{-1}, corresponding to the B–O bond vibration in trigonal BO_3 groups.[44] It should be

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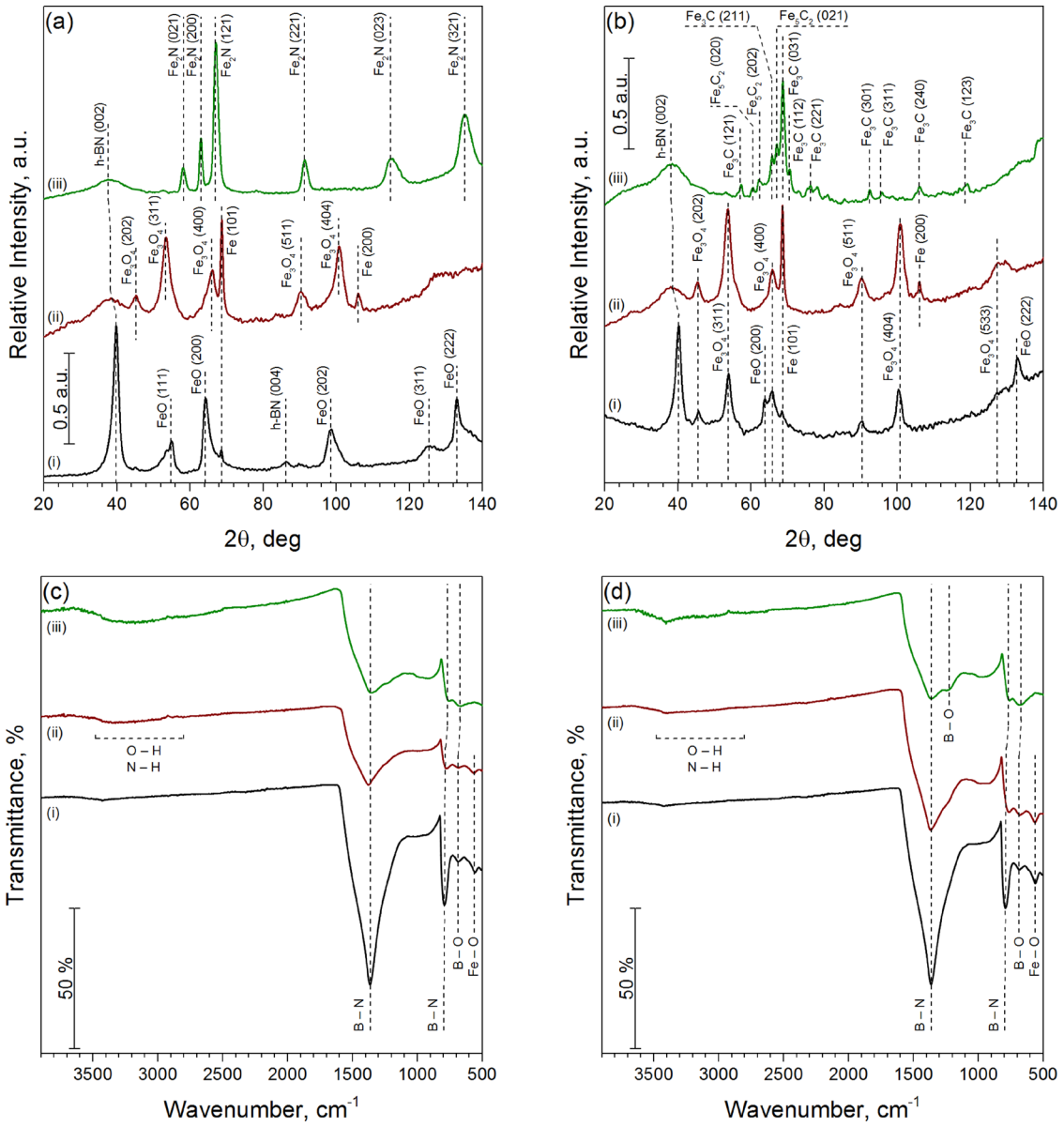


Figure 2. XRD patterns (a,b) and FTIR spectra (c,d) of Fe/*h*-BN (i), Fe/*h*-BN-300 (ii), and Fe/*h*-BN-500 (iii) samples before (a,c) and after (b,d) activation at 230 °C in a CO_2/H_2 mixture overnight.

noted that the relative intensities of the O−H and N−H bonds remained unchanged after treatment, which indicates the thermal stability of these bonds upon activation.

X-ray photoelectron spectroscopy (XPS) survey spectra of the Fe/*h*-BN nanohybrids are presented in Figure S1. The surface of samples consists of Fe (712 eV), O (532 eV), N (398 eV), and B (190 eV) elements. Note that there is no Cl in the samples, indicating that iron chloride is a suitable precursor for hydroxide-mediated deposition of iron on the support surface. The elemental compositions derived from the XPS spectra are presented in Table 1. The results show that the N:B ratio decreases as the intensity of mechanical treatment increases, which can be caused by the loss of nitrogen from the *h*-BN.[28] The decrease of oxygen content in the sample obtained at 500 rpm should also be noted. It is well known that the iron surface is prone to oxidation while iron nitrides are more oxidation-resistant.[45] Thus, the decrease of oxygen content in the sample can be attributed to the formation of iron nitride, which is also confirmed by the XRD analysis.

Table 1. Elemental Composition of the Fe/*h*-BN Surface

sample	element content, at %			
	Fe	O	N	B
Fe/*h*-BN	1.60	59.33	16.14	22.92
Fe/*h*-BN-300	2.52	57.42	15.95	24.10
Fe/*h*-BN-500	1.31	40.68	14.82	43.19

High-resolution XPS spectra of the Fe 2p, N 1s, and B 1s peaks are shown in Figure 3. The XPS Fe 2p spectra showed doublets at 713 and 726.2 eV corresponding to the Fe $2p_{3/2}$ and Fe $2p_{1/2}$ states, respectively.[46] The Fe $2p_{3/2}$ peak of the Fe/h-BN sample produced by H_2 reduction was deconvoluted into two components centered at 712.7 and 710.5 eV, corresponding to iron in +3 and +2 oxidation states, respectively.[47] The iron oxides can be present either as separate phases or as the passivation layer on the surface of metallic iron. After ball milling of the Fe/*h*-BN sample at 300 rpm, the intensity of the peak corresponding to +3 iron state

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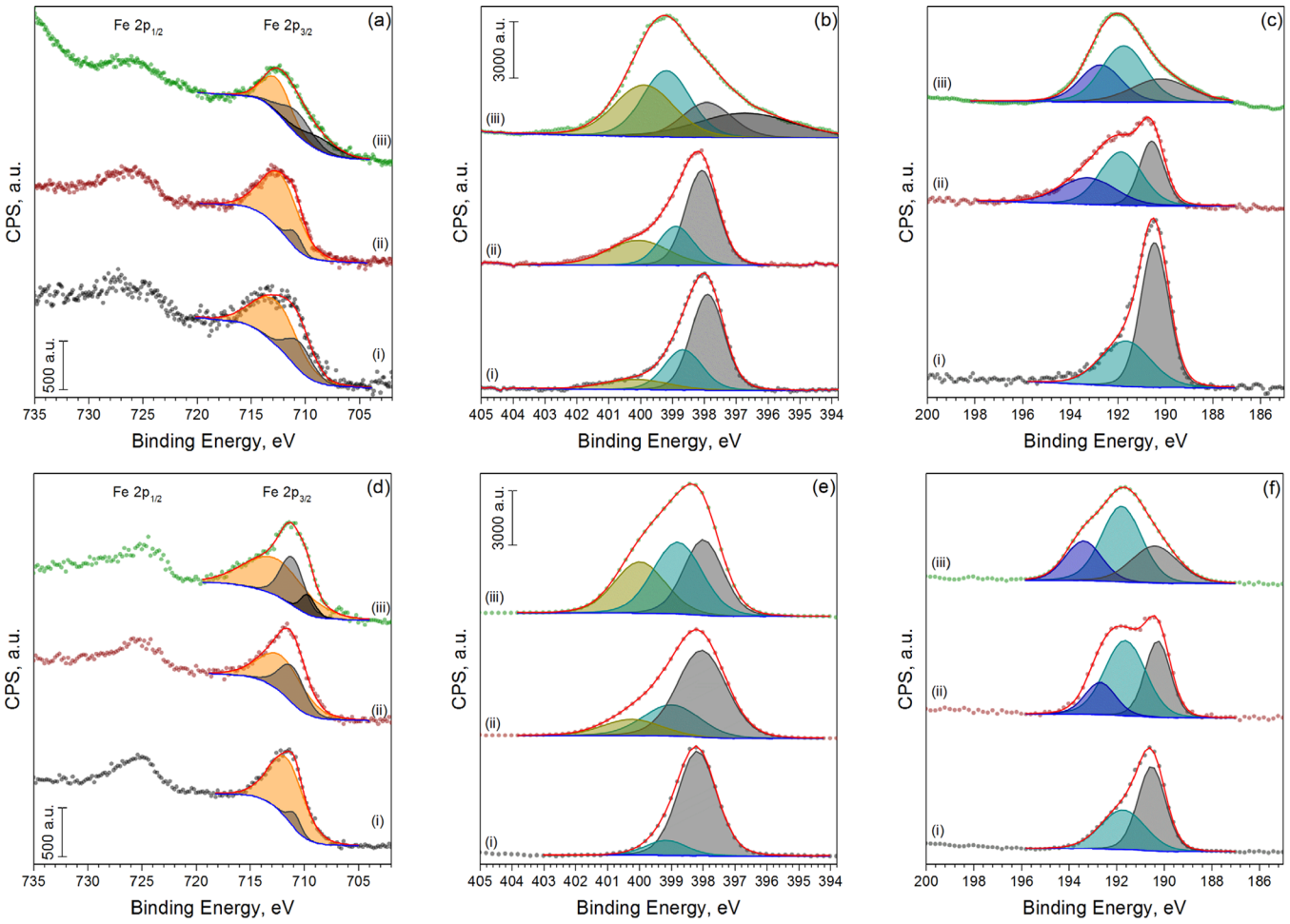


Figure 3. High-resolution XPS Fe 2p (a,d), N 1s (b,e), and B 1s (c,f) spectra of Fe/*h*-BN (i), Fe/*h*-BN-300 (ii), and Fe/*h*-BN-500 (iii) samples before (a–c) and after (d–f) activation at 230 °C in the CO_2/H_2 mixture overnight.

increased. This agrees well with the formation of a magnetite phase observed in the corresponding XRD pattern. Further increase in grinding speed led to broadening of the Fe 2p peak. Deconvolution of the peak into three components at 712.7, 710.8, and 708.8 eV shows the presence of the Fe^{3+}, Fe^{2+}, and Fe in the form of nitride.[48]

The N 1s peak of the sample produced via H_2 reduction was deconvoluted into three components centered at 397.9, 398.7, and 400.1 eV corresponding to nitrogen surrounded by boron atoms,[41] nitrogen bonded with boron and oxygen,[49] and nitrogen in the B–N–H environment.[41] An increase in the intensity of the 400.1 eV peak indicates an increase in the contribution from the B–N–H groups. This is also consistent with the FTIR spectroscopy results suggesting the formation of N–H bonds. A further increase in the ball-milling speed leads to the appearance of a new component at 396.7 eV corresponding to nitrogen in iron nitride.[50]

The XPS B 1s spectrum of the hydrogen-reduced sample showed two features at 190.5 and 191.7 eV, which were ascribed to the BN and BNO components, respectively.[44] The mechanical treatment led to the formation of a new peak at 193.3 eV corresponding to boron in the oxygen environment.[51] After more intensive treatment, a redistribution of the peak intensity between the components is observed. This confirms the FTIR spectroscopy results on the surface functionalization with a hydroxyl group.

After activation, an increase in the intensity of Fe^{3+} peaks is observed, indicating that an increased content of ions in the oxide layer formed after exposure of the samples to air. Only the Fe/*h*-BN-500 sample shows the presence of a low-energy peak at 709.2 eV, which can be attributed to Fe in carbide, which is consistent with XRD analysis. Deconvolution of the high-resolution N 1s and B 1s XPS peaks after activation shows a decrease in the amount of BNO component.

Morphology, Size, and Structure of Synthesized Fe/*h*-BN Catalysts. Transmission electron microscopy (TEM) and high-angle annular dark-field scanning transmission electron microscopy (HAADF STEM) images, as well as Fe particle size distribution and Fe elemental maps, are presented in Figure 4. The Fe/*h*-BN sample mostly consists of the Fe-based needle-shaped particles uniformly distributed over the *h*-BN surface. The ball milling changes the particle morphology from needle-like to spherical or irregular shaped. An interesting observation is that the average size of Fe particles is not changed during mechanical processing and remains equal to 10 nm, although the size distribution becomes wider.

The high-resolution transmission electron microscopy (HRTEM) image presented in Figure S2a reveals a characteristic fringe contrast originated from the (101) Fe planes with an interplanar distance (d) of 0.20 nm, (202), (311), and (404) Fe_3O_4 planes with a d spacing of 0.30, 0.25, and 0.15 nm, respectively, and (002) *h*-BN planes with a *d* spacing of

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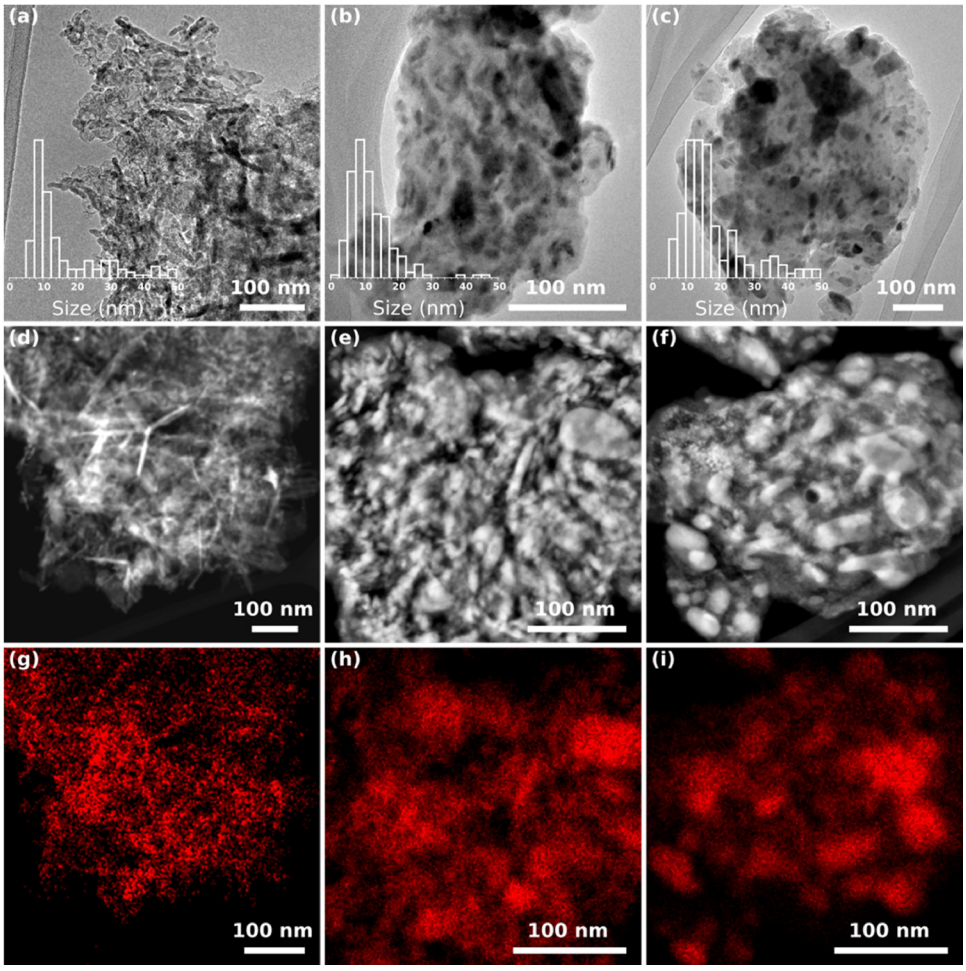


Figure 4. TEM (a–c) and HAADF STEM (d–f) images, Fe particle size distribution (insets in a–c), and Fe elemental maps (g–i) of Fe/*h*-BN (a,d,g), Fe/*h*-BN-300 (b,e,h), and Fe/*h*-BN-500 (c,f,i) samples.

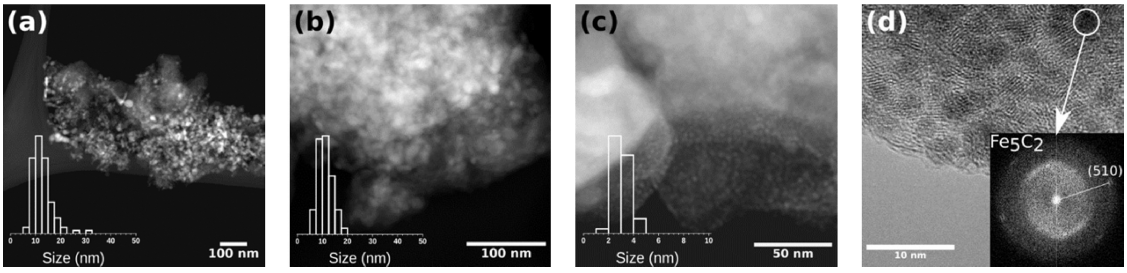


Figure 5. HAADF STEM (a–c) and HRTEM (d) images of Fe/*h*-BN (a), Fe/*h*-BN-300 (b), and Fe/*h*-BN-500 (c,d) after activation at 230 °C in a CO_2/H_2 mixture overnight.

0.35 nm, as was calculated from the corresponding fast Fourier transform (FFT) images. High defectiveness of the *h*-BN after the mechanical treatment at 300 rpm is evidenced by the halo on the FFT pattern (Figure S2b). The HRTEM image reveals nanoparticles, 10 nm in size, located on the BN support surface. A characteristic distance of 0.20 nm between the (101) planes indicates that this is Fe nanoparticle. After the treatment at 500 rpm, two specific features can be noted: (i) the *h*-BN fringes completely disappeared in the corresponding HRTEM image and (ii) the Fe_2N nanoparticle was identified based on the characteristic interplanar distance between the (121) planes in the corresponding FFT pattern.

After activation, the number of needle-like nanoparticles in the Fe/*h*-BN sample decreased (Figure 5a). Taking into account the change in the phase composition, this correlates with a decrease in the Fe(II) oxide content. The particle morphology in the Fe/*h*-BN-300 sample does not change during activation, and the nanoparticle size distribution remains the same as after BM with an average size of 10 nm (Figure 5b). In the case of the Fe/*h*-BN-500 sample, many fine nanoparticles 2–4 nm in size are formed after activation (Figure 5c). This correlates well with the formation of iron carbide in this sample and is further confirmed by the corresponding FFT pattern in the HRTEM image (Figure 5d, inset).

Adsorption Characteristics of Fe/*h*-BN Catalysts. After ball milling, the Brunauer–Emmett–Teller (BET) specific surface area of the Fe/*h*-BN catalysts decreased from 213 to 14

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m^2/g (Table 2). A similar effect was observed during the processing of micron *h*-BN powder in a ball mill.[41] As the treatment time increased, the specific surface area decreased from 404 to 155 m^2/g. Because *h*-BN nanopowder with a large surface area (Ssp = 213 m^2/g) was used as the starting material in the present study, the decrease in the surface area is explained by particle agglomeration.

Table 2. Adsorption Characteristics of Fe/*h*-BN Samples

sample	BET specific surface area m^2/g	CO_2 adsorption $\mu mol/g$	H_2 adsorption $\mu mol/g$	CO adsorption $\mu mol/g$
Fe/*h*-BN	213	1.1	4.1	0.852
Fe/*h*-BN-300	21	3.6	2.6	0.063
Fe/*h*-BN-500	14	4.7	3.3	4.8

Hydrogen, carbon dioxide, and carbon monoxide chemisorption strongly depended on the ball-milling speed of Fe/*h*-BN samples. Weight-normalized CO_2 adsorption increased after ball milling by a factor of 5. Although weight-normalized hydrogen chemisorption decreased after the treatment, one should note that area-normalized adsorption is actually increased from 19 up to 236 $nmol/m^2$. Thus, the total surface concentration of active centers capable of activating CO_2 and H_2 increased after the ball milling. To better understand the origin of active centers, ball-milled *h*-BN powder at 300 and 500 rpm was also studied in CO_2 and H_2 adsorption experiments. Hydrogen adsorption is negligible in the case of untreated *h*-BN powder, and a slight increase in adsorption is observed after mechanical treatment (Table S1). Note that

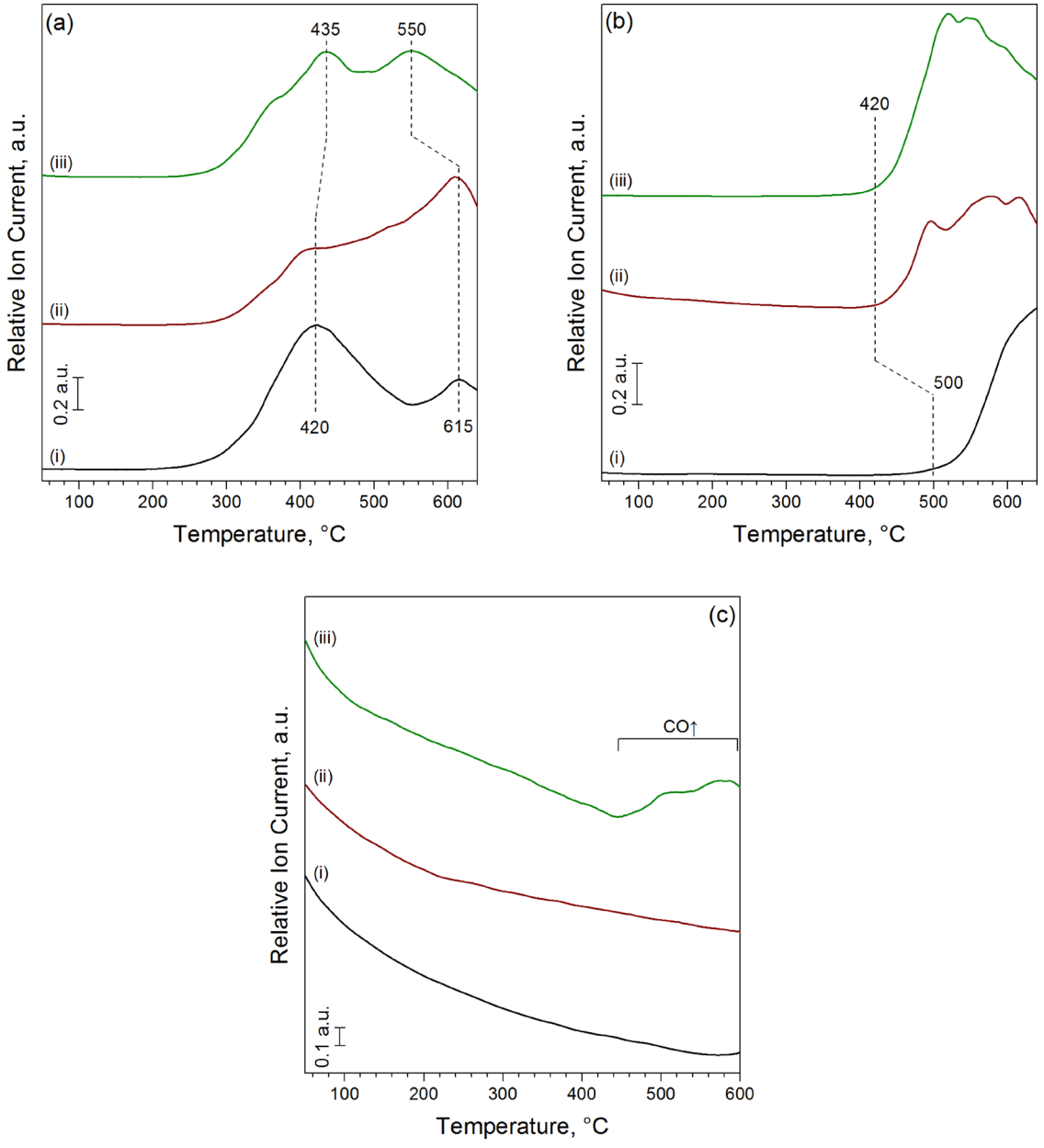


Figure 6. TPD curves for CO_2 (a), H_2 (b) and CO (c). (i) Fe/*h*-BN, (ii) Fe/*h*-BN-300, and (iii) Fe/*h*-BN-500. Abscissa on the plots is the ratio of the ion current to its maximum over the entire temperature range.

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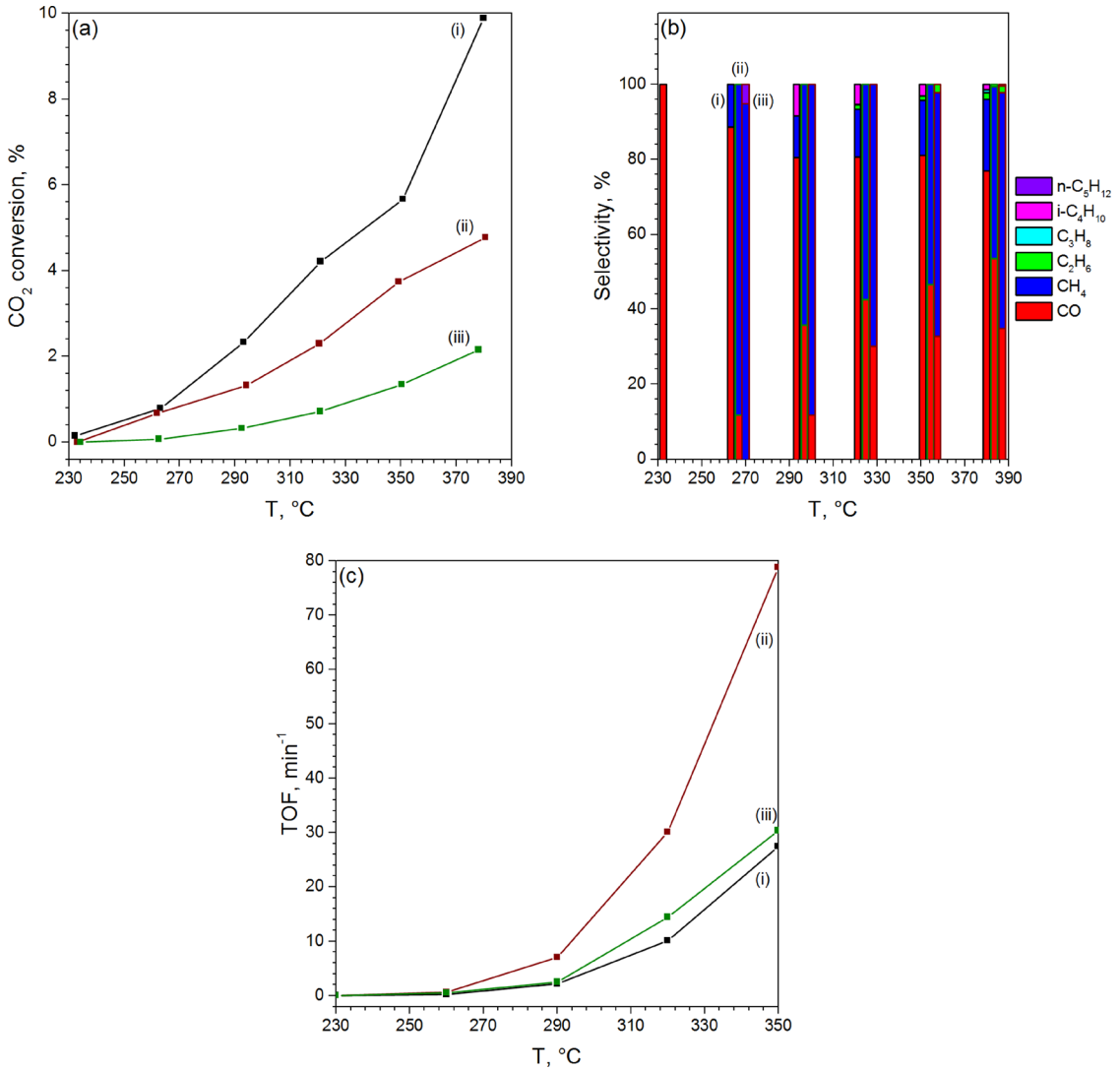


Figure 7. CO_2 conversion (a), selectivity (b), and TOF (c) of Fe/*h*-BN (i), Fe/*h*-BN-300 (ii), and Fe/*h*-BN-500 (iii) catalysts.

adsorption is at least an order of magnitude lower than that of nanohybrids, which indicates that Fe-based phases are the main hydrogen activation centers in nanohybrids. The CO_2 chemisorption on *h*-BN and nanohybrids does not differ significantly. Chemisorption increases after ball milling, indicating the formation of CO_2 activation centers on the BN surface (Table S1). Interesting behavior demonstrates CO chemisorption. The amount of adsorbed CO was negligible in the case of Fe/*h*-BN and Fe/*h*-BN-300 samples while it underwent a steep increase in the case of the Fe/*h*-BN-500 sample.

Temperature-programmed desorption (TPD) curves for the Fe/*h*-BN, Fe/*h*-BN-300, and Fe/*h*-BN-500 samples are presented in Figure 6. When using the Fe/*h*-BN sample, the CO_2 TPD measurements reveal two desorption peaks at 420 and 615 °C. After ball milling with a rate of 300 rpm, the positions of peaks remain unchanged; however, the first peak almost vanishes, while the intensity of the second increases significantly. A further increase in the ball-milling rate leads to a 65° shift of the high-temperature desorption peak toward a lower value (Figure 6a). Similar results were obtained for Fe-free *h*-BN powder. Characteristic peaks of CO_2 desorption at 450 and 535−550 °C are shown in Figure S3.

When using the Fe/*h*-BN sample, the H_2 desorption starts at 500 °C and does not finally complete up to 650 °C. After mechanical treatment, the temperature of the onset of H_2 desorption decreases by 80 °C, and the reaction proceeds in the temperature range from 420 to >640 °C. The CO desorption data are consistent with adsorption and show no TPD peaks for Fe/*h*-BN and Fe/*h*-BN-300 samples. The Fe/*h*-BN-500 catalyst shows a CO desorption peak above 450 °C.

CO_2 Hydrogenation. CO_2 conversion and selectivity of the Fe/*h*-BN catalysts were measured in integral mode, that is, at high catalyst loading and low reactant total flow rate, and the results are presented in Figure 7. It can be seen that CO_2 conversion decreases after the sample mechanical treatment in a ball mill (Figure 7a). Note that because the specific surface area of samples is also decreased after ball milling (Table 2), the decrease in conversion can originate from the samples' total surface area decrease.

Selectivity with respect to various hydrogenation products was determined in the integral mode of the plug-flow reactor. It is clearly seen that mechanical treatment has a significant effect on selectivity at different temperatures. At 230 °C, only the Fe/*h*-BN sample has non-zero conversion with 100% selectivity to CO. The Fe/*h*-BN sample shows almost 90%

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selectivity to CO at 260 °C. After ball-milling, there is a selectivity shift toward HCs, with the Fe/*h*-BN-500 sample showing 100% selectivity at the same temperature. However, with an increase in temperature, the selectivity to HCs increases only in the case of the Fe/*h*-BN nanomaterial and decreases for the samples after mechanical treatment. Nevertheless, at 380 °C, the highest HC yield is observed for the Fe/*h*-BN-500 nanohybrids, which indicates the efficiency of processing in a ball mill. Among HCs, the highest selectivity for all catalysts is observed for methane; however, up to 10% of the formation of HCs C_{2-5} was observed for the Fe/*h*-BN and Fe/*h*-BN-500 nanomaterials.

CO_2 conversion is an integral characteristic that is affected by the catalyst surface area and measurement conditions such as flow rates and catalyst loading. To assess the intrinsic activity of samples, the turnover frequency (TOF) values were calculated as the ratio of the CO_2 hydrogenation rate determined in the differential mode (i.e., at low catalyst loading and a high total reactant flow rate) to the number of active sites (determined by H_2 chemisorption). Figure 7c shows that the TOF values increased significantly after the ball milling at 300 rpm and decreased after the processing at an increased speed.

Characteristics of Samples after Catalytic Hydrogenation. Catalytic CO_2 hydrogenation using the Fe/*h*-BN nanocomposites led to irreversible changes in their phase composition and surface state. The results of XRD analysis showed that, after catalysis, the FeO phase was completely transformed into magnetite and metallic iron (Figure S4a). According to XPS data, the surface of iron-containing nanoparticles contains Fe^0, Fe^{2+}, and Fe^{3+} ions (Figure S4b). The *h*-BN became less crystalline, as evidenced by the increase in the FWHM value of the (002) peak. This can be explained by the partial oxidation of *h*-BN by water vapors formed during the hydrogenation process. The increased amount of B−N−O groups (Figure S4c,d) further confirms this assumption. There are no carbide phases in the sample after catalysis, which explains its low selectivity with respect to HCs compared to other samples. The formation of the Fe_5C_2 phase was observed in the Fe/*h*-BN-300 sample after catalysis (Figure S4a). Thus, the ability of the catalyst to activate the CO_2 molecule due to the *h*-BN surface functional groups formed as a result of mechanical treatment contributed to the formation of carbide active phases during catalysis. However, it should be noted that this transformation was not completed during the tests, and iron oxide was still present in the sample after catalysis. The Fe/*h*-BN-500 sample, which after activation contained iron carbides and showed the highest selectivity toward HCs, also revealed carbides after catalysis. However, the XRD peaks became sharper, indicating an increase in crystallite sizes. In the Fe/*h*-BN-300 and Fe/*h*-BN-500 samples, the *h*-BN did not change after catalysis and its surface still contains B−N−O and B−N−H groups (Figure S4c,d), which are necessary for the activation of the CO_2 molecule.

The morphology and size of the Fe nanoparticles in the Fe/*h*-BN and Fe/*h*-BN-300 samples did not change after the catalytic experiments (Figure S5a,b). This is not the case for the Fe/*h*-BN-500 sample, which suffers from noticeable agglomeration of iron carbide particles with a 10-fold increase in size (Figure S5c). HRTEM images of the nanocomposites confirm the presence of magnetite and carbide nanoparticles in the untreated and ball-milled samples, respectively (Figure S5d−f).

DISCUSSION

Influence of Mechanical Treatment on the Characteristics of Fe/*h*-BN Catalysts. The XRD analysis shows that during mechanical treatment, the interaction between the Fe-containing phases and *h*-BN has occurred. Because the treatment was carried out in a protective Ar atmosphere and after mechanical processing, the samples were passivated in O_2, the only nitrogen source for the formation of iron nitride is BN. The observed increase in the (002) *h*-BN interplanar spacing and broadening of its diffraction peaks can be attributed to the increased imperfection of lattice caused by the corresponding nitrogen loss.[28] Taking into account the decrease in the intensity of iron oxide peaks, the formation of iron nitride can occur either as a result of the reaction between metallic iron and BN, or by a metathesis reaction between iron oxide and BN:

$$2Fe + BN = Fe_2N + B \quad (1)$$

$$8FeO + 4BN = 4Fe_2N + 2B_2O_3 + O_2 \quad (2)$$

Both reactions are endergonic (unfavorable) under standard conditions (1 bar total pressure and activity of reactants equal to 1) with free Gibbs energies of 235 and 519 kJ/mol for reactions 1 and 2, respectively, at 298 K. This indicates the need for additional energy, which can be provided by ball milling.[52] Reaction 2 is less probable because it requires 2.2 times more energy than reaction 1. Additional arguments in favor of reaction 1 are that the formation of boron oxide phases in the XRD patterns and an increase in the B−O peak intensity in the FTIR spectra are not observed. The disappearance of XRD FeO peaks can be explained by disproportionation reaction 3, which is exergonic with a Gibbs energy of −34 kJ/mol:

$$4FeO = Fe_3O_4 + Fe \quad (3)$$

The formation of iron nitride can proceed directly by diffusion reaction 1 as proposed by Guo et al.[53] during ball milling of Fe and *h*-BN powder mixtures. The second possible route is the stepwise formation of iron nitride by reactions 4 and 5:

$$BN = B + 1/2N_2 \quad (4)$$

$$2Fe + 1/2N_2 = Fe_2N \quad (5)$$

Reactions 4 and 5 are also endergonic with standard free Gibbs energies of 227 and 8 kJ/mol, respectively. Under nonstandard conditions (i.e., at low partial pressure of nitrogen in the system), the Gibbs energy of reaction 4 can become negative. This is true in our case because we use an Ar protective atmosphere for mechanical treatment. Although the Gibbs energy of reaction 5 is positive, it requires much less energy compared to direct reaction 1. Therefore, this reaction appears to be more likely in the mechanical strain field caused by the high-energy ball-milling process. The formation of iron nitride from iron in a nitrogen atmosphere was explained by the "ballistic" diffusion mechanism.[54,55]

After ball milling, surface functionalization of Fe/*h*-BN samples with hydroxyl and amino groups is observed, as evidenced by the results of FTIR and XPS analysis. Because it is unlikely that this occurs during the BM stage in a protective Ar atmosphere, it is reasonable to assume that the presence of BN lattice defects leads to the formation of terminated B−O−

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H and B−N−H bonds when the samples are exposed to air as a result of reaction with moisture:[41]

$$BN + H_2O \rightarrow B - OH + B - NH \quad (6)$$

The following scheme (Figure 8) of phase transformations occurring during the mechanical treatment of Fe/*h*-BN

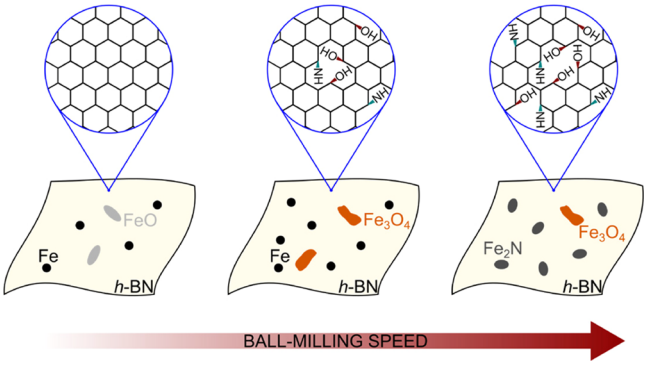


Figure 8. Scheme of phase transformations occurring during mechanical treatment of Fe/*h*-BN nanocomposites.

nanocomposites can be proposed. The initial sample is *h*-BN nanosheets decorated with iron and iron oxide nanoparticles. BM in an Ar atmosphere leads to the formation of a large number of defects in the BN_x structure ($x < 1$) and simultaneous disproportionation of ferrous oxide with the formation of Fe and Fe_3O_4 phases. An increase in the BM rate leads to an increase in the system energy, which becomes sufficient for the nitriding of iron with nitrogen formed after the partial decomposition of BN. After passivation in O_2/Ar flow, the newly formed iron-containing phases are covered with a thin oxide layer. The defective BN surface is not completely passivated, leaving unsaturated bonds, which are compensated for by reaction with water vapor when exposed to ambient air. This leads to the surface functionalization by hydroxyl and amino groups, which can act as acid and base sites, increasing the adhesion of CO_2 molecules during hydrogenation.

Effect of Activation in a CO_2/H_2 Mixture on Fe/*h*-BN Nanocomposites. Activation in a CO_2/H_2 mixture has a different effect on the Fe/*h*-BN, Fe/*h*-BN-300, and Fe/*h*-BN-500 nanocomposites. An untreated Fe/*h*-BN sample undergoes phase composition changes with a decrease in the FeO oxide content and a simultaneous increase in the magnetite and Fe^0 phases. Because iron(II) oxide is unstable at the activation temperature, it most likely decomposes according to reaction 3 during the activation process. The *h*-BN microstructure does not change during activation in this sample. However, a decrease in the content of functional B−N−H groups is observed, which indicates a slight surface oxidation. The morphology of iron-containing nanoparticles changes during activation with a decrease in the number of needle-like particles and the formation of spherical ones, which can be explained by a change in the sample phase composition.

A sample subjected to moderate BM oxidizes during or immediately after the activation, which is accompanied by an increase in the content of the magnetite phase. The formation of iron oxide after activation can be explained by the oxidation of metallic iron in air. The structure of *h*-BN is not changed after activation, although analysis of the surface composition shows a slight increase in the content of B−O−H groups compared to the nonactivated sample. The morphology and size of iron-containing nanoparticles do not change during activation, which indicates their structural stability provided by the nanoparticles-*h*-BN support interaction.

The most interesting changes occur in the activated Fe/*h*-BN-500 sample. The nitride phase disappears after exposure to the CO_2/H_2 mixture, but new carbide phases Fe_3C and Fe_2C_5 appear. It is well established that Hägg iron carbide plays an important role in the CO_2 hydrogenation reaction, being an active phase for the growth of a HC chain.[56]

Because the iron nitride phase serves as a precursor for the iron carbide phase, it increases the selectivity toward HC formation. Calculation of the formation energies of iron carbide from different precursors shows that thermodynamically, the pathway mediated by iron nitride is more favorable in comparison with metallic iron. The iron carbide formation reaction from iron shows a more endothermic character compared to the iron nitride in both CO and CO_2 environments.

$$5Fe_2N + 4CO = 2Fe_5C_2 + 5/2N_2 + 2O_2 \quad \Delta E = 0.13 \text{ eV}$$

$$5Fe_2N + 4CO_2 = 2Fe_5C_2 + 5/2N_2 + 4O_2 \quad \Delta E = 1.36 \text{ eV}$$

$$5Fe + 2CO = Fe_5C_2 + O_2 \quad \Delta E = 0.36 \text{ eV}$$

$$5Fe + 2CO_2 = Fe_5C_2 + 2O_2 \quad \Delta E = 1.59 \text{ eV}$$

However, when hydrogen is added to the reaction mixture, the reaction type changes to an exothermic one for both environments, although the reaction is more exothermic when the iron nitride transforms to iron carbide.

$$5Fe_2N + 4CO + 4H_2 = 2Fe_5C_2 + 5/2N_2 + 4H_2O \quad \Delta E = -0.82 \text{ eV}$$

$$5Fe_2N + 4CO_2 + 4H_2 = 2Fe_5C_2 + 5/2N_2 + 8H_2O \quad \Delta E = -0.55 \text{ eV}$$

$$5Fe + 2CO + 2H_2 = Fe_5C_2 + 2H_2O \quad \Delta E = -0.59 \text{ eV}$$

$$5Fe + 2CO_2 + 4H_2 = Fe_5C_2 + 4H_2O \quad \Delta E = -0.32 \text{ eV}$$

Thus, the presence of hydrogen in the reaction mixture plays a key role in the formation of iron carbide. Moreover, the mechanism of the most exothermic reaction involves the participation of CO and hydrogen with the formation of iron carbide, nitrogen, and water as reaction products. The transition from iron nitride to iron carbide was observed during CO_2 hydrogenation over unsupported iron nitride.[57]

CO_2 Hydrogenation Reaction Using Ball-Milled Fe/*h*-BN Nanocomposites. The CO_2 reduction reaction on the Fe/*h*-BN, Fe/*h*-BN-300, and Fe/*h*-BN-500 nanocomposites proceeds differently. The BM at 300 rpm leads to an increase in the TOF. This result is in a good agreement with an increase in CO_2 and H_2 adsorption showing the increase of the active site amount after the BM processing. TPD data show that there are two types of active centers for CO_2 activation on the sample surface. It is well known that the base centers can activate CO_2 molecules.[58] Moreover, Lewis base centers, such as nitrogen in the B−N−H species, were reported for defective *h*-BN.[41] According to the XPS N1s spectrum (Figure 3e), the Fe/*h*-BN sample contains 12 and 29% of B−N−H and B−N−O groups, respectively. The mechanical treatment increases the number of B−N−H groups up to 25%, which implies an increase in the number of strong CO_2 adsorption centers. In the Fe-free *h*-BN sample after mechanical treatment, two types of CO_2 activation centers are also observed. When the ball-milling speed is increased, the number of strong sites increases (Figure S3a). Thus, two types of CO_2 adsorption centers on

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the defective BN surface are attributed to the B−N−O and B−N−H groups responsible for weak and strong sites, respectively.

Further increase in the BM rate leads to a decrease in TOF values, accompanied by a simultaneous increase in selectivity to HCs. The adsorption studies have showed that after BM, the number of CO_2 and H_2 activation centers increases, which is consistent with increased BN defectiveness. At first glance, the decrease in activity is in contradiction with the increase in the number of active sites. However, one should consider the increased adsorption of CO molecules for the Fe/*h*-BN-500 sample. On the one hand, stronger interaction between CO and the catalyst surface leads to the increase in HC selectivity. On the other hand, there is a competition between the reaction components for adsorption sites, which leads to a decrease in the overall activity. Available data indicate that the main difference in the sample catalytic activity is the presence of iron carbide in the activated sample which is responsible for the increased selectivity toward HCs.

The calculated values of CO adsorption energy are presented in Table S2 and are between −1.06 and −1.83 eV (Fe), −1.17 and −1.62 eV (Fe_2N), and −0.62 and −1.54 eV (Fe_5C_2) for various surface sites. The most energetically favorable positions of CO adsorption are shown in Figure S6. Despite the fact that the CO adsorption energy on iron carbide is somewhat lower compared to metallic iron, the C−O bond length is maximum when it is adsorbed on the Fe_5C_2 surface (1.22 Å). An increase in the bond length significantly reduces the CO activation energy and promotes the further formation of HCs.[59]

Based on the experimental results, the following catalytic mechanism for mechanically treated Fe/*h*-BN nanomaterials has been proposed. After BM, subsequent passivation in an O_2/Ar gas flow, and further exposure to air, the sample surface is functionalized with hydroxyl and amino groups. B−N−O and B−N−H surface species are Lewis base centers capable of activating CO_2 and effectively increasing the activity of the catalyst in CO_2 hydrogenation. However, because of the high activation energy of CO molecules on the catalyst surface, CO is the main product formed during the reaction. An increase in the BM rate leads to an increase in the system energy, and, as a consequence, to an increase in the number of CO_2 activation centers. At the same time, iron nitride, which is formed at the activation stage as a result of Fe-BN interaction, is transformed into a mixture of iron carbides, which facilitate CO molecule activation. As a result, the selectivity with respect to HCs is increased because of enhanced CO reduction kinetics via the Fischer−Tropsch process. At the same time, the intrinsic activity decreases because of the competition of reactant molecules for adsorption sites.

CONCLUSIONS

Fe/*h*-BN nanocomposites with a functionalized surface were obtained by the precipitation method and were subjected to subsequent BM in an Ar atmosphere, further passivation in O_2/Ar flow, and final exposure to air. The as-obtained nanohybrids were characterized as promising catalysts in the CO_2 hydrogenation reaction. The most important results are summarized as follows.

The Fe/*h*-BN sample consists of ferrous oxide, metallic iron, and *h*-BN. Mechanical treatment at a rate of 300 rpm leads to disproportionation of ferrous oxide with the formation of iron and magnetite, as well as to *h*-BN lattice defects due to the loss of nitrogen. Functionalization of the *h*-BN surface with hydroxyl and amino groups occurs as a result of the interaction of surface defects with water vapor from ambient air. After mechanical processing with a rate of 500 rpm, the formation of iron nitride nanoparticles is observed. Thermodynamic analysis of the reactions shows that the most probable mechanism for the iron nitride formation is the direct nitriding of metallic iron with nitrogen formed after the partial decomposition of *h*-BN.

The increased catalytic activity after BM at a rate of 300 rpm is attributed to the formation of Lewis basic sites on the *h*-BN surface. The main product of CO_2 hydrogenation is CO with a selectivity of 92.8%, which is related to the high activation energy of CO molecules on the catalyst surface. The increase in the processing rate up to 500 rpm leads to higher selectivity toward the formation of HCs, which is associated with the iron nitride → carbide transformation during the activation stage. An increase in selectivity leads to a decrease in catalytic activity due to the competition of reactant molecules for adsorption sites. Density functional theory (DFT) calculations show a significant weakening of the CO bond on the iron carbide surface and that the mechanism for the formation of Fe_5C_2 from Fe_2N is more favorable.

MATERIALS AND METHODS

Materials. *h*-BN nanosheets (Plazmoterm, Russia) were annealed under vacuum at 1450 °C overnight prior to use. Iron chloride tetrahydrate ($FeCl_2{\cdot}4H_2O$, Reahim, Russia) and sodium hydroxide (NaOH, Reahim, Russia) were of analytical grade and used without further purification.

Synthesis of Fe/*h*-BN Nanohybrids. The synthesis of Fe/*h*-BN nanohybrids was performed via an iron hydroxide-mediated route.[60] 2.67 g of $FeCl_2{\cdot}4H_2O$ were dissolved in 1.5 L of 10 vol % aqueous ethanol solution. Then, 3 g of *h*-BN were ultrasonically dispersed in the solution for 30 min. After sonication, 800 mL of 2 g/L NaOH aqueous solution was added dropwise to the dispersion with vigorous stirring at room temperature for 8 h. The final pH of supernatant solution was equal to 9. The precipitates were washed with distilled water by centrifugation to neutral pH and finally dried under ambient conditions overnight.

The iron hydroxide/boron nitride powder was heat treated in an H_2 atmosphere to obtain Fe/*h*-BN nanohybrids. For this, 2.5 g of sample were heated at a rate of 10 °C/min in 400 mL/min H_2 flow and held at a final temperature of 500 °C for 4 h. Then, the powder was cooled down to room temperature in H_2 flow and passivated in 1 vol % O_2/Ar flow overnight.

Ball Milling of Fe/*h*-BN Nanohybrids. The Fe/*h*-BN nanohybrids were ball-milled in a Fritsch Pulverisette 7 planetary micro mill at different grinding speeds. 600 mg of sample were placed in zirconia vials with six zirconia 10 mm balls. The samples were sealed in an Ar atmosphere in a glovebox and ground at 300 and 500 rpm for 8 h in start-stop mode (5 min rotation and then 5 min pause). These ball-milling speeds were chosen based on the results of preliminary studies, because at a speed >500 rpm, contamination of the nanohybrids with zirconia was observed. After the mechanical treatment, samples were passivated in 1 vol % O_2/Ar flow to avoid uncontrollable oxidation. Hereafter, samples reduced in an H_2 atmosphere at 500 °C and then BM at 300 and 500 rpm are denoted as Fe/*h*-BN-300 and Fe/*h*-BN-500, respectively.

Material Characterization. The phase composition was analyzed on a Difrey-402 diffractometer (Scientific instruments, Russia) equipped with a positional sensitive detector using Cr-Kα irradiation (λ = 2.29106 Å) in a 2θ range from 20 to 140°. The morphology of nanohybrids and the size and distribution of catalytically active nanoparticles on the support surface were studied using transmission electron microscopy (TEM) on a JEM-2100 instrument (Jeol, Japan) equipped with an energy-dispersive spectroscopy detector. The specific surface area of powders was measured by the low-temperature

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nitrogen adsorption method on a NOVA 1200e instrument (Quantachrome, USA), and the obtained results were processed using the BET equation. Prior to adsorption measurements, the samples were degassed under vacuum at 200 °C overnight. The surface chemical state was analyzed using XPS and FTIR spectroscopy methods. XPS spectra were obtained on an Axis Supra spectrometer (Shimadzu, Japan). The samples after the ball-milling treatment, activation in a mixture of H_2/CO_2, and after the catalysis were analyzed. A fitting process was conducted with CasaXPS software after the subtraction of Shirley-type background. The C1s peak (284.8 eV) was used to calibrate XPS spectra. FTIR spectra were recorded on a Vertex 70v instrument (Bruker, Germany) in a 500−4000 cm^{-1} range with 4 cm^{-1} resolution. The samples after the ball-milling treatment and activation in a mixture of H_2/CO_2 gases were analyzed. Iron content in the synthesized materials was measured using inductively coupled plasma optical emission spectroscopy using an ICAP-6300 Radial spectrometer (Thermo Scientific, USA). Iron from the nanohybrids was dissolved in aqua regia for the analysis. The iron content was measured to be 28 ± 2 wt % for all samples.

Temperature-Programmed Desorption Studies. CO_2, H_2, and CO TPD and pulse titration studies were carried out on a ChemBET Pulsar instrument (Quantachrome, USA). For the analysis, 100 mg of sample were mixed with quartz granules and placed in a U-shaped quartz tube. The samples were treated in a CO_2/H_2 gas mixture at 400 °C for 4 h and then annealed in an He flow at the same temperature for the next 4 h. After the treatment, the samples were cooled to room temperature, after which, using He as a transport gas, a dose of adsorbate gas (150 μL per pulse) was injected into the system. The gas composition at the outlet of the reactor was monitored by a quadrupole mass spectrometer Thermostat GSD 320 (Pfeiffer vacuum, Germany). The amount of adsorbed gas was calculated as the difference between the average peak area after saturation of the catalyst surface and the peak areas during the adsorption process, calibrated using a known amount of gas. After adsorption, the samples were heated to 650 °C at a rate of 5 °/min in a He flow. The effluent gases were analyzed with a mass spectrometer.

CO_2 Hydrogenation. Sample intrinsic catalytic activity was measured in a fixed-bed continuous-flow reactor in differential mode, that is, at low CO_2 conversion (no more than 5% of the maximum theoretical value). In a typical experiment, 2 mg of a sample was mixed with quartz granules and placed in a stainless steel reactor with an inner diameter of 1.5 mm. Prior to analysis, the samples were activated in a gas flow of CO_2 (6 mL/min) and H_2 (12 mL/min) at 230 °C overnight. The hydrogenation activity was studied at 20 atm in a flow of 6 mL/min CO_2 and 12 mL/min H_2 (VHSV = 540 L·g_{cat}^{-1}·h^{-1}) in a temperature range from 230 to 350 °C. The product composition was measured with a Crystal 5000 online chromatograph (Chromatek, Russia) equipped with three thermal conductivity detectors, a flame ionization detector, packing columns M ss316 NaX 80/100 mesh 2000 mm × 2 mm^2, HayeSep R 80/100 mesh 1000 mm × 2 mm^2, M ss316 HayeSep Q 80/100 mesh 2000 mm × 2 mm^2, and a capillary column MXT—Alumina BOND/MAPD 30000 mm × 0.53 mm^2.

The catalyst intrinsic activity was calculated as TOF values:

$$\mathrm{TOF} = \frac{f_{i,CO_2} - f_{f,CO_2}}{m_{cat} \cdot n_{H_2}} \quad (7)$$

where $f_{i,CO2}$ − molar CO_2 flow at the inlet of the reactor, mol/min; $f_{f,CO2}$ − molar CO_2 flow at the outlet of the reactor, mol/min; m_{cat} − catalyst weight, g; n_{H2} − H_2 adsorption, mol/g.

CO_2 conversion and catalyst selectivity to different reaction products were measured in a plug-flow reactor in integral mode. For these experiments, 100 mg of catalyst granules (200−315 μm) were mixed with quartz granules up to 1 mL volume and placed into the reactor between quartz wool plugs. Activation and measurement regimes were similar to the intrinsic activity determination experiments.

CO_2 conversion was calculated as follows:

$$\mathrm{conversion} = \frac{f_{i,CO_2} - f_{f,CO_2}}{f_{i,CO_2}} \quad (8)$$

where $f_{i,CO2}$ − molar CO_2 flow at the inlet of the reactor, mol/h; $f_{f,}CO_2$ − molar CO_2 flow at the outlet of the reactor, mol/h.

Catalyst selectivity for various reaction products was calculated as follows:

$$s_j = \frac{\nu_j X_j}{\sum \nu_j X_j} \quad (9)$$

where s_j is selectivity toward the jth product; ν_j is the stoichiometric coefficient in the chemical reaction with the jth product; and X_j is the concentration of the jth product at the outlet of the reactor.

The carbon balance was calculated as a ratio of carbon in converted CO_2 and in the reaction products. The results were accepted if carbon balance was more than 98%.

Computational Methods. All quantum-chemical calculations were performed within the framework of DFT[61,62] using the plane-wave basis set and the projector augmented wave method[63] as implemented in the VASP program package.[64−66] The revised Perdew, Burke, and Ernzerhof exchange-correlation functional was used.[67] The plane wave basis cutoff energy E_{cutoff} = 400 eV was used. The Grimme corrections[68] were applied in order to take into account the van der Waals interactions. The Brillouin zone was integrated using Gamma center grids with a constant spacing density of 0.2 $Å^{-1}$ transverse to the slabs. Relaxation of the atomic geometry was carried out until the maximum values of the energy difference between two electronic steps became less than 10^{-4} eV, and the energy variation between the two steps of the electronic loop became less than 10^{-5} eV. To avoid interactions between periodic images, we used a vacuum region of at least 15 Å in nonperiodic directions. The Fe slab was modeled from *bcc* iron in the (111) direction with a supercell size of 7.04 and 8.11 Å in the lateral direction after relaxation. The Fe_2N slab was constructed from an e-Fe_2N unit cell[69] in the (010) direction with a supercell size of 10.15 and 9.07 Å in the lateral direction after relaxation. The Fe_5C_2 slab was constructed from a monoclinic C2/c-Fe_5C_2 unit cell[70] in the (100) direction with a supercell size of 8.99 and 9.98 Å in the lateral direction after relaxation.

The formation energy was calculated as the difference between the energies of the products and reactants per iron atom according to the equation:

$$\Delta E = \frac{E(\mathrm{Products}) - E(\mathrm{Reactants})}{nFe}$$

The energies of Fe, Fe_2N, and Fe_5C_2 were obtained from DFT calculations for the corresponding crystals, and the energies of CO, CO_2, N_2, O_2, H_2, and H_2O were obtained from calculations of isolated molecules.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsanm.2c03540.

XPS survey spectra; HRTEM images of Fe/*h*-BN, Fe/*h*-BN-300, and Fe/*h*-BN-500 samples; TPD curves for CO_2 and H_2; XRD, high-resolution XPS, HAADF STEM, and HRTEM of samples after catalytic hydrogenation of CO_2; schematics of CO molecule adsorption on different surfaces; adsorption characteristics of *h*-BN samples; and sorption energies and C-O bond length calculated via DFT (PDF)

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The manuscript was written through contributions of all the authors. All the authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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